

START

OPEN the accident records CSV file

READ all accident records

STORE the records as a list of dictionaries

CHECK the unique accident causes

STANDARDIZE the accident causes

 REMOVE extra spaces

 CAPITALIZE the first letter of each word

CONVERT vehicles_involved to integer

CONVERT casualties to integer

CREATE an empty list called valid_records

CREATE an empty list called invalid_records

FOR each accident record

 CREATE an empty list called reasons

 IF vehicles involved is less than 0

 ADD "Vehicles involved cannot be negative" to reasons

 END IF

 IF casualties is less than 0

 ADD "Casualties cannot be negative" to reasons

 END IF

 IF severity is not Minor, Serious or Fatal

 ADD "Invalid severity" to reasons

 END IF

 IF location is empty

 ADD "Location is missing" to reasons

 END IF

 IF cause is empty

 ADD "Cause is missing" to reasons

 END IF

 IF time period is empty

 ADD "Time period is missing" to reasons

 END IF

 IF reasons is empty

 ADD record to valid_records

 ELSE

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        ADD record and reasons to invalid_records
    END IF

END FOR

CALCULATE total accidents
    COUNT the valid records

CALCULATE total casualties
    ADD casualties from each valid record

COUNT accidents by severity

COUNT accidents by cause

IDENTIFY the most common cause

COUNT accidents by location

CALCULATE total casualties for each location

IDENTIFY location or locations with most accidents

IDENTIFY location or locations with most casualties

COUNT accidents by time period

IDENTIFY time period or periods with most accidents

COUNT fatal accidents

CALCULATE percentage of fatal accidents

DEFINE function to view accident records

DEFINE function to analyse severity

DEFINE function to analyse causes

DEFINE function to analyse locations and time periods

DEFINE function to view invalid records

DEFINE function to display safety summary

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DISPLAY menu

WHILE user has not selected Exit

DISPLAY:

1. View accident records
2. Analyse severity
3. Analyse causes
4. Analyse locations and time periods
5. View invalid records
6. View safety summary
7. Exit

GET user's choice

IF choice is 1

CALL view accident records function

ELSE IF choice is 2

CALL analyse severity function

ELSE IF choice is 3

CALL analyse causes function

ELSE IF choice is 4

CALL analyse locations and time periods function

ELSE IF choice is 5

CALL view invalid records function

ELSE IF choice is 6

CALL safety summary function

ELSE IF choice is 7

DISPLAY "Exiting the program"

BREAK the loop

ELSE

DISPLAY "Invalid selection Choose a number between 1 and 7"

RETURN to the menu

END IF

END WHILE

END